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On the treewidth of toroidal grids[☆]Masashi Kiyomi^a, Yoshio Okamoto^b, Yota Otachi^{c,*}^a International College of Arts and Sciences, Yokohama City University, Seto 22-2, Kanazawa-ku, Yokohama, Kanagawa, 236-0027, Japan^b Department of Communication Engineering and Informatics, Graduate School of Informatics and Engineering, The University of Electro-Communications, Chofugaoka 1-5-1, Chofu, Tokyo 182-8585, Japan^c School of Information Science, Japan Advanced Institute of Science and Technology, Asahidai 1-1, Nomi, Ishikawa 923-1292, Japan

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ABSTRACT

Many graph parameters of grid-like graphs are studied because of their algorithmic consequences. An important concept in this context is that of treewidth. Treewidth of graphs is a graph parameter for measuring how close a graph is to a tree. In this paper, we study the treewidth of toroidal grids and show that the treewidth of the $n \times n$ toroidal grid is either $2n - 2$ or $2n - 1$. We then show that these bounds are tight in some cases. To show the lower bounds, we construct brambles of high orders.

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1. Introduction

Treewidth of graphs is one of the most successful concepts for measuring how close a graph is to a tree. This concept was introduced by Robertson and Seymour [9] in their project on graph minor theory. It is known that if input graphs have bounded treewidth, then many generally-hard problems become polynomial-time solvable (see e.g., [1]). In the study of treewidth, *square grid graphs* (or just *grids*) play important roles. The $n \times n$ grid has treewidth exactly n [2], and thus a graph with an $n \times n$ grid minor has treewidth at least n . Furthermore, it is known that a graph has bounded treewidth if and only if the size of a maximum grid minor in the graph is bounded [10].

In this paper, we investigate the treewidth of a *toroidal grid*, i.e., a grid with the wrap around edges between the top-most row and the bottom most row and between the left-most column and the right-most column. Some graph parameters related to treewidth have been studied for such grid-like graphs. For example, these parameters include pathwidth [3], bandwidth [8], and carving-width [5]. Recently, Wood [12] has shown that the treewidth of the $n \times n$ toroidal grid is at least $2n - 5$ and at most $2n$. Improving these bounds, we show that the treewidth of the $n \times n$ toroidal grid is $2n - 2$ or $2n - 1$, and that these bounds are tight.

2. Preliminaries

Let G be a graph. We denote by $V(G)$ and $E(G)$ the vertex set and the edge set of G , respectively. Let X and Y be subsets of $V(G)$. If the subgraph of G induced by X is connected, then we say that X is *connected*. Two vertex sets X and Y *touch* if they

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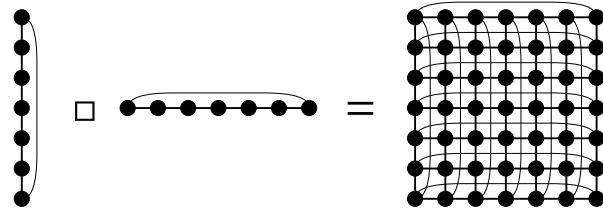


Fig. 1. The 7×7 toroidal grid T_7 .

intersect or there is an edge between them. A subset family $\mathcal{S} = \{S_1, \dots, S_k\}$ of $V(G)$ is a *bramble* of G if every subset $S_i \in \mathcal{S}$ is connected and any two subsets in $\mathcal{S}$ touch. The *order* of $\mathcal{S}$ is the size of a minimum *hitting set* of $\mathcal{S}$; that is, the minimum size of a subset $H \subseteq V(G)$ that intersects every subset $S_i \in \mathcal{S}$. If H is a hitting set of $\mathcal{S}$ and $S_i \in \mathcal{S}$, then we call an element $h \in H \cap S_i$ a *hitting vertex* of S_i . The *bramble number* of a graph G , denoted by $\mathbf{bn}(G)$, is the maximum order over all brambles of G .

A *tree-decomposition* of a graph G is a pair $(\mathcal{X}, T)$, where T is a tree and $\mathcal{X} = \{X_i \mid i \in V(T)\}$ is a collection of subsets of $V(G)$ such that:

- $\bigcup_{i \in V(T)} X_i = V(G)$;
- for each edge $\{v, w\} \in E(G)$, there is a node $i \in V(T)$ such that $v, w \in X_i$;
- for each $v \in V(G)$, the set of nodes $\{i \mid v \in X_i\}$ forms a subtree of T .

The *width* of a tree-decomposition $(\mathcal{X}, T)$ is $\max_{i \in V(T)} |X_i| - 1$. The *treewidth* of G , denoted by $\mathbf{tw}(G)$, is the minimum width over all tree-decompositions of G . A *path-decomposition* of G is a tree-decomposition $(\mathcal{X}, T)$ in which T is a path. The *pathwidth* of G , denoted by $\mathbf{pw}(G)$, is the minimum width over all path-decompositions of G . Clearly, $\mathbf{tw}(G) \leq \mathbf{pw}(G)$ for any graph G . Seymour and Thomas [11] showed that for any graph G , it holds that

$$\mathbf{tw}(G) = \mathbf{bn}(G) - 1.$$

By this fact, we can show a lower bound of treewidth by constructing a bramble of high order. Actually, in this paper, our main contribution is to construct a bramble of the $n \times n$ toroidal grid, and to show that the bramble has high order.

Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be graphs. The *Cartesian product* $G_1 \square G_2$ is the graph whose vertex set is $V_1 \times V_2$, and any two vertices (u_1, v_1) and (u_2, v_2) are adjacent in $G_1 \square G_2$ if and only if either $v_1 = v_2$ and $\{u_1, u_2\} \in E_1$, or $u_1 = u_2$ and $\{v_1, v_2\} \in E_2$. We denote by C_n the cycle of n vertices $\{1, 2, \dots, n\}$ and n edges $\{\{1, 2\}, \{2, 3\}, \dots, \{n-1, n\}, \{n, 1\}\}$. The $n \times n$ toroidal grid T_n is the Cartesian product $C_n \square C_n$ of two copies of the cycle C_n (see Fig. 1).

Wood [12] showed that the treewidth of T_n is at least $2n - 5$. Ellis and Warren [3] showed that the pathwidth of T_n is $2n - 1$. Combining these results, we have $2n - 5 \leq \mathbf{tw}(T_n) \leq 2n - 1$. We will narrow down this gap by showing that $\mathbf{tw}(T_n) \geq 2n - 2$. Note that the general lower bound of $\mathbf{tw}(G \square H)$ due to Kozawa et al. [6] gives only a constant lower bound for toroidal grids T_n .

3. The lower bound

Here we show that $\mathbf{tw}(T_n) \geq 2n - 2$ and thus have the theorem below. We also show the tightness of the bounds by showing that $\mathbf{tw}(T_n) = 2n - 1$ if $n = 3$ and $\mathbf{tw}(T_n) = 2n - 2$ if $n = 4$.

Theorem 3.1. $2n - 2 \leq \mathbf{tw}(T_n) \leq 2n - 1$ for any $n \geq 3$.

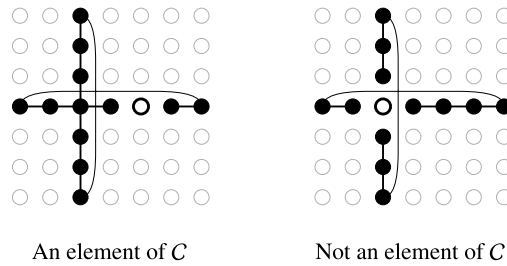
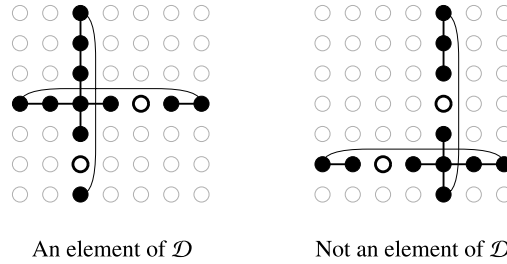
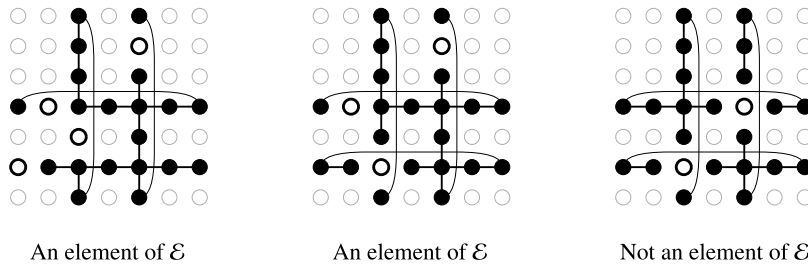
The row i of T_n is the vertex set $\{(i, j) \mid 1 \leq j \leq n\}$, and the column j of T_n is the vertex set $\{(i, j) \mid 1 \leq i \leq n\}$. To show the lower bound, we construct a bramble $\mathcal{B} = \mathcal{C} \cup \mathcal{D} \cup \mathcal{E}$ as follows (see Figs. 2–4).

- An element of $\mathcal{C}$ is obtained from the union of one row and one column by removing one vertex, where the removed vertex is not the intersection vertex of the row and the column. The family $\mathcal{C}$ consists of all such sets.
- An element of $\mathcal{D}$ is obtained from the union of one row and one column by removing one vertex from the row and one vertex from the column, where the vertex removed from the column is strictly below the row and the vertex removed from the row is not the intersection vertex.
- An element of $\mathcal{E}$ is obtained from the union of two rows and two columns by removing one vertex from each row and from each column, where at most one of the removed vertices may be an intersection vertex of a row and a column. The family $\mathcal{E}$ consists of all such sets.

We call a vertex set obtained by removing at most one vertex from a row or from a column, respectively, a *semi-row* or a *semi-column*. We say that semi-rows and semi-columns without missing vertices are *complete*.

Observation 3.2. A semi-row i and a semi-column j touch if at least one of them contains the vertex (i, j) . $\square$

Note that, from the definitions above, every element of $\mathcal{B}$ includes at least one semi-row and at least one semi-column.

Fig. 2. Family $\mathcal{C}$.Fig. 3. Family $\mathcal{D}$.Fig. 4. Family $\mathcal{E}$.

Lemma 3.3. The family $\mathcal{B} = \mathcal{C} \cup \mathcal{D} \cup \mathcal{E}$ is a bramble of T_n .

Proof. From the definition, every element of $\mathcal{B}$ is connected. Let $B_1, B_2 \in \mathcal{B}$.

From the definitions, each element of $\mathcal{B}$ includes semi-row and semi-column. This fact and [Observation 3.2](#) together imply that B_1 and B_2 touch if at least one of them is an element of $\mathcal{C}$ because each element of $\mathcal{C}$ includes a complete row or a complete column.

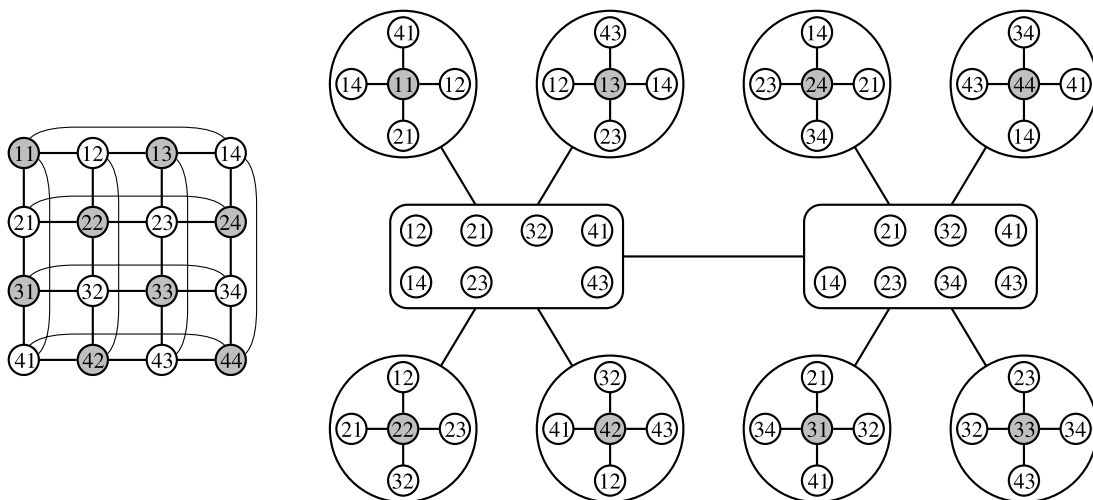
Now assume that one, say B_1 , is in $\mathcal{E}$ and the other, B_2 , is in $\mathcal{D} \cup \mathcal{E}$. Let B_1 include semi-rows i and i' , and let B_2 include a semi-column j . From the definition of semi-columns, the semi-column j contains at least one of the vertices (i, j) and (i', j) , and thus, by [Observation 3.2](#), it touches the semi-row i or i' . Consequently, B_1 and B_2 touch.

In the remaining case, we have $B_1, B_2 \in \mathcal{D}$. Let B_1 include a semi-row i and a semi-column j , and let B_2 include a semi-row i' and a semi-column j' . Assume without loss of generality that $i \leq i'$. By the definition of B_2 , the missing vertex in the semi-column j' is below the row i' and hence (i, j') is in B_2 . By [Observation 3.2](#), B_1 and B_2 touch. $\square$

Lemma 3.4. The order of $\mathcal{B}$ is at least $2n - 1$.

Proof. Suppose that $\mathcal{B}$ has a hitting set H of size at most $2n - 2$. Since $|H| < 2n$, there are a row R and a column C such that $|H \cap R| \leq 1$ and $|H \cap C| \leq 1$. If $H \cap R = \emptyset$ or $H \cap C = \emptyset$, then there is a set $A \in \mathcal{C}$, with $A \subset R \cup C$, that has empty intersection with H . This contradicts the assumption that H is a hitting set of $\mathcal{B} \supseteq \mathcal{C}$, and thus H intersects all rows and columns. Now, by the pigeon hole principle, there are two rows i and i' and two columns j and j' such that each has exactly one hitting vertex. Without loss of generality, we may assume that $i < i'$.

Let $S = (R_i \cup R_{i'} \cup C_j \cup C_{j'}) \setminus H$ and $M = (R_i \cup R_{i'} \cup C_j \cup C_{j'}) \cap H$, where R_k and C_k denote the sets of vertices in the row k and the column k , respectively. If at most one of the vertices in M is an intersection vertex of the rows and the columns, then $S \in \mathcal{E}$. However, this cannot happen since $S \cap H = \emptyset$ and H is a hitting set of $\mathcal{E}$. Thus at least two vertices of M are intersection vertices of the rows and the columns. On the other hand, if M contains three or more vertices at intersections, it must contain two vertices in one of the rows and the columns. This contradicts the choice of S . We thus conclude that M contains exactly two vertices and they are at intersections. Without loss of generality, assume that $M = \{(i, j'), (i', j)\}$.

Fig. 5. A tree-decomposition of T_4 .

Let T be the set of vertices obtained from the union of the row i and the column j by removing the vertices (i, j') and (i', j) . From the definition, $T \in \mathcal{D}$. Since $T \subset S$ and $S \cap H = \emptyset$, it follows that $T \cap H = \emptyset$. This is a contradiction because H is a hitting set of $\mathcal{D}$. $\square$

Now we discuss the tightness of the bounds in Theorem 3.1. Fig. 5 shows that $\text{tw}(T_4) \leq 6$. Thus the lower bound $2n - 2$ in Theorem 3.1 is tight for $n = 4$. Note that the decomposition in Fig. 5 cannot be generalized for larger n . Lucena [7] showed that $\text{tw}(K_n \square K_n) = n^2/2 + n/2 - 1$. Therefore, $\text{tw}(T_3) = 5$ since $C_3 = K_3$. This implies that the upper bound $2n - 1$ in Theorem 3.1 is tight for $n = 3$.

4. Concluding remarks

We have shown that $\text{tw}(T_n) \in \{2n - 2, 2n - 1\}$ and that these bounds are tight. The problem of deciding which bound applies for each n remains unsettled. It is known that the $h \times w$ grid $P_h \square P_w$ has treewidth exactly $\min\{h, w\}$ [2], where P_n is the path of n vertices $\{1, 2, \dots, n\}$ and $n - 1$ edges $\{1, 2\}, \{2, 3\}, \dots, \{n - 1, n\}$. Since the $h \times w$ cylinder $P_h \square C_w$ contains $P_h \square P_w$ as a minor, the treewidth of $P_h \square C_w$ is at least $\min\{h, w\}$. Since $\text{tw}(P_h \square C_w) \leq \text{pw}(P_h \square C_w) = \min\{2h, w\}$ [3], it holds that

$$\min\{h, w\} \leq \text{tw}(P_h \square C_w) \leq \min\{2h, w\}.$$

Since $\text{tw}(C_h \square C_w) \leq \text{pw}(C_h \square C_w) \leq 2 \cdot \min\{h, w\}$ [3] and $C_h \square C_w$ contains $C_{\min\{h, w\}} \square C_{\min\{h, w\}}$ as a minor, our result in this paper implies that

$$2 \cdot \min\{h, w\} - 2 \leq \text{tw}(C_h \square C_w) \leq 2 \cdot \min\{h, w\}.$$

Closing the gaps above would be nice future work. Note that tree-decompositions (actually path-decompositions) attaining the upper bounds can be obtained as described by combining the results of Kinnersley [4] and Ellis and Warren [3].

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